

CO3-AT1-Debugging and Optimization

Exp-1:

```
2
3 public class Debug1{
4     public static void main(String args[]){
5         Scanner sc=new Scanner(System.in);
6         int arr[]={20,40,60,80};
7         if(sc.hasNextInt()){
8             int index=sc.nextInt();
9             if(sc.hasNextInt()){
10                 int divisor=sc.nextInt();
11                 try{
12                     try{
13                         int value=arr[index];
14                         int result=value/divisor;
15                         System.out.println(result);
16                     }
17                     catch(ArrayIndexOutOfBoundsException e){
18                         System.out.println("Invalid Index");
19                     }
20                     catch(ArithmeticException e){
21                         System.out.println("Division by Zero");
22                     }
23                 }catch(Exception e){
24                     System.out.println("An error occurred");
25                 }
26             }
27         }
28         sc.close();
29     }
30 }
```

0

Output

Division by zero

Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

Exp 2:

Your Fixed Code

```
1 import java.util.Scanner;
2
3 public class Calculator {
4     static int divide(int a, int b) {
5         return a / b;
6     }
7     static void display(int x) {
8         System.out.println("Answer = " + x);
9     }
10    public static void main(String[] args) {
11        Scanner sc = new Scanner(System.in);
12        int a = sc.nextInt();
13        int b = sc.nextInt();
14        try {
15            display(divide(a, b));
16        }
17        catch (ArithmeticException e) {
18            System.out.println("Cannot Divide by Zero");
19        }
20        finally {
21            System.out.println("Execution Completed");
22        }
23        sc.close();
24    }
25 }
```

Test Input

36
6

Output

Answer = 6
Execution Completed

Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

Exp 3:

Your Fixed Code

```
1 import java.util.Scanner;
2
3 public class Debug3 {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         int arr[] = {12, 24, 36, 48};
8         String number = sc.next();
9
10        try {
11            int index = Integer.parseInt(number);
12            try {
13                System.out.println(arr[index]);
14            } catch (ArrayIndexOutOfBoundsException e) {
15                System.out.println("Invalid Array Index");
16            }
17        } catch (NumberFormatException e) {
18            System.out.println("Invalid Number");
19        }
20    }
21 }
```

Test Input

5

Output

Invalid Array Index

Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

✓ Submitted

Exp 4:

Your Fixed Code

```
1 import java.util.Scanner;
2 public class Main {
3     static int calculate(int a, int b) {
4         try {
5             return a / b;
6         } catch (ArithmeticException e) {
7             return 0;
8         } finally {
9             System.out.println("Resources Released");
10        }
11    }
12    public static void main(String[] args) {
13        Scanner sc = new Scanner(System.in);
14        int a = sc.nextInt();
15        int b = sc.nextInt();
16        if (b == 0) {
17            calculate(a, b);
18            System.out.println("Division by Zero");
19        } else {
20            int result = calculate(a, b);
21            System.out.println(result);
22        }
23        sc.close();
24    }
25 }
```

Test Input

20 5

Output

```
Resources Released
4
```

Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks

Exp 5:

Your Fixed Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String name = sc.nextLine();
6         if (name.equalsIgnoreCase("Null")) {
7             System.out.println("String is Null");
8         } else {
9             System.out.println("Length = " + name.length());
10        }
11        sc.close();
12    }
13 }
```

Test Input

Java

Output

```
Length = 4
```

Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks

✓ Submitted